

UNIT 6

SOIL

Learning objectives:

- Identify the components of soil (minerals, organic matter, water, air, and living organisms) and explain their roles in supporting plant growth.
- Differentiate between the types of soil (sandy, clayey, and loamy) based on their properties and uses in agriculture.
- Describe the soil profile, including the layers (topsoil, subsoil, and bedrock), and understand their composition and functions.
- Recognize the importance of soil in supporting plant and animal life, filtering water, and preventing erosion.

SOIL

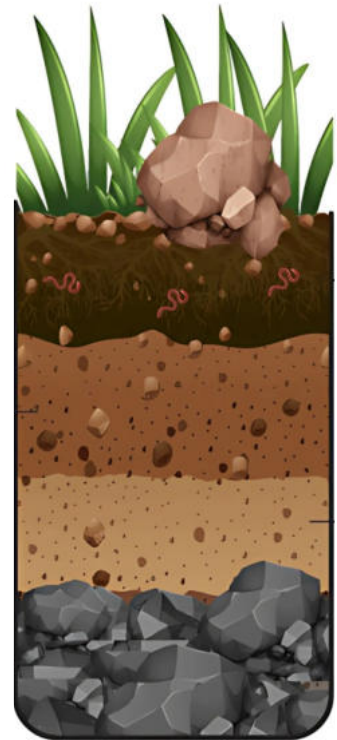
Soil is one of the most important natural resources on Earth. It provides nutrients to plants, supports animal life, and helps store water. Soil forms over thousands of years through the breakdown of rocks (weathering) and the decomposition of organic matter like dead plants and animals. Different types of soil have different properties, which makes them suitable for various plants and crops.



COMPONENTS OF SOIL

Soil is made up of four main components:

- **Minerals:** These are tiny rock particles that come from the breakdown of larger rocks.
- **Organic Matter:** Decayed plants and animals that add nutrients to the soil.
- **Water and Air:** Fill the spaces between soil particles, helping plants grow.
- **Living Organisms:** Earthworms, bacteria, and insects that improve the soil's quality.



TYPES OF SOIL

Sandy Soil:

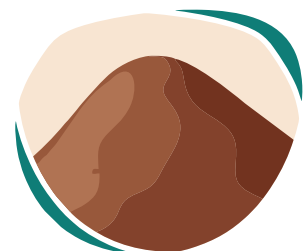
- Large particles.
- Drains water quickly.
- Poor in nutrients, making it less fertile.

Clayey Soil:

- Made of tiny particles that stick together.
- Holds water well but is dense and heavy.
- Fertile but hard to plow.

Loamy Soil:

- A mix of sand, silt, and clay.
- Drains water well but retains enough moisture for plants.
- Rich in nutrients, making it ideal for agriculture.



LAYERS OF SOIL (SOIL PROFILE)

Soil is arranged in layers called a **soil profile**.

Topsoil:

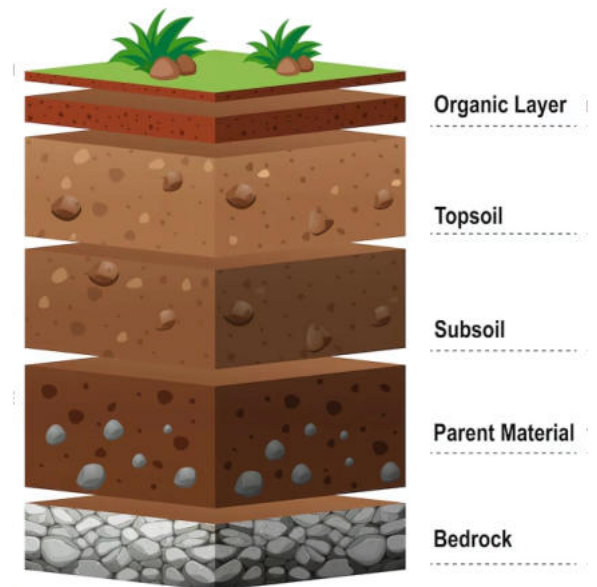
- The topmost layer, rich in organic matter.
- Plants grow here.

Subsoil:

- Below the topsoil, it contains minerals like iron and clay.
- Fewer organic materials than the topsoil.

Bedrock:

- The bottom layer, made of solid rock.
- It eventually breaks down to form soil.



IMPORTANCE OF SOIL

- **Supports Plant Life:** Provides nutrients, water, and a base for roots.
- **Habitat for Animals:** Home to many insects, worms, and small animals.
- **Filters Water:** Removes impurities as water passes through.
- **Prevents Erosion:** Holds the land together and stops it from washing away.



Fill in the blanks.

1. Soil is made of minerals, organic matter, _____ and _____.
2. The top layer of soil where plants grow is called _____.
3. _____ soil holds water but is dense and sticky.
4. Loamy soil is a mix of sand, _____ and clay.
5. _____ is the layer of solid rock beneath the soil.
6. Earthworms and bacteria in the soil improve its _____.
7. Sandy soil drains _____ quickly but is less fertile.
8. _____ soil is ideal for growing crops because it retains nutrients and water.



Write “T” for True or “F” for False.

1. Topsoil is the layer where most plant roots grow.

2. Sandy soil makes soil less fertile.

3. Clayey soil drains water faster than sandy soil.

4. Soil contains living organisms like earthworms.

5. Bedrock is found above the topsoil.

6. Loamy soil is suitable for agriculture because it retains nutrients and moisture.

7. Soil is formed in just a few years.

8. Subsoil contains fewer nutrients than topsoil.





Choose the correct answer.

1. Which soil type drains water the fastest?

- Sandy soil
 - Clayey soil
 - Loamy soil
 - Topsoil
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2. What does topsoil contain?

- Only minerals
 - Organic matter and nutrients
 - Solid rocks
 - Air only
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3. Which soil type is ideal for farming?

- Sandy soil
 - Clayey soil
 - Loamy soil
 - Rocky soil
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4. Soil is made up of:

- Rocks and air
 - Organic matter, minerals, water, and air
 - Only sand and water
 - Living organisms only
-





5. What is bedrock?

- A type of soil
 - The bottom layer of soil, made of solid rock
 - A mix of sand and clay
 - The layer where plants grow
-

6. Clayey soil is best suited for:

- Growing crops
 - Making pottery
 - Building houses
 - All of the above
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7. Which type of soil holds the most water?

- Sandy soil
- Clayey soil
- Loamy soil
- Subsoil



Answer the following questions.

Q1: What are the four main components of soil?

Q2: Why is topsoil important for plants?

Q3: What makes loamy soil better for farming compared to sandy or clayey soil?

Q4: What is the role of water and air in the soil?

Q5: Name the three layers of soil in a soil profile and describe.

Q6: How do earthworms help improve the soil?



Critical Thinking

What would happen if all topsoil on Earth disappeared? How would it affect plants animals, and humans?

- Encourage students to consider the role of soil in agriculture, water storage, and the ecosystem.





Testing Soil Types



ACTIVITY

Understand the properties of sandy, clayey, and loamy soils.

- **Materials Needed:** Samples of each soil type, water, jars, and a worksheet.

- **Instructions:**

1. Place a small amount of each soil type in separate jars.
2. Add water to each jar and shake well.
3. Let the soil settle and observe the layers formed.
4. Record which soil holds water the longest and which drains the fastest.

- **Outcome:** Students will identify soil types and understand their properties.

